

Weather Prediction Model for Wenzhou Kean University Using Random Forest and Holt-Winters

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Abstract: This study develops a weather prediction framework for Wenzhou Kean University (28.0000°N, 120.7000°E), combining machine learning and time series smoothing. A Random Forest Regression (RFR) model is first trained on 172,634 samples of hourly meteorological data—including temperature, solar radiation, and precipitation—to generate short-term hourly temperature forecasts. To enhance seasonal consistency and reduce volatility, the hourly predictions are subsequently adjusted using Holt-Winters Exponential Smoothing, which is fitted separately to historical daily temperature data from 1979 to 2018. Model performance is evaluated using R^2 , MAE, and RMSE, achieving $R^2 = 0.94$, MAE = 0.18, and RMSE = 0.26. While RFR captures nonlinear patterns in hourly features, the Holt-Winters model, operating at the daily level, improves alignment with historical seasonal trends. Limitations regarding the decoupled modeling structure and reliance on seasonal assumptions are discussed. This two-stage approach provides a practical solution for high-resolution, campus-level temperature forecasting.

Keywords: Hourly temperature forecasting, Random Forest Regression, Holt-Winters Exponential Smoothing, Time Series Analysis, Seasonal Adjustment

1. Introduction to Weather Prediction

Weather prediction is a vital tool for institutional planning, resource management, and safety assurance. This project develops a forecasting model for Wenzhou Kean University (28.0000°N, 120.7000°E) on May 14, 2025, using a Random Forest Regression model (RFR) and Seasonal Decomposition and Holt-Winters Exponential Smoothing model (Holt-Winters). The RFR model is used to predict hourly temperatures, while the Holt-Winters model optimizes these predictions by adjusting for seasonal trends. Both models achieve high accuracy, supporting applications such as campus energy management, event planning, and operational scheduling.

1.1. How It Works

This project integrates two complementary forecasting approaches. The Random Forest Regression (RFR) model applies an ensemble of decision trees, each trained on random subsets of features and samples. By averaging predictions across trees, it captures complex, nonlinear relationships in tabular weather data for robust hourly forecasts.

In parallel, the Holt-Winters model uses seasonal decomposition to split temperature series into trend, seasonal, and residual components. Exponential smoothing is then applied to each, enabling accurate daily forecasts by modeling recurring patterns and long-term trends.

1.2. Key Concepts

- **Random Forest Regression (RFR):** An ensemble method that combines multiple decision trees to improve prediction accuracy and reduce overfitting [1].
- **Seasonal Decomposition and Holt-Winters Exponential Smoothing (Holt-Winters):** A time series forecasting method that decomposes data into trend, seasonal, and residual components, applying exponential smoothing to capture recurring patterns and long-term trends [2, 3].
- **Evaluation Metrics:** R^2 , Mean Absolute Error (MAE), and Root Mean Square Error (RMSE) are used to quantify model performance for both RFR and Holt-Winters.
- **Feature Importance:** RFR identifies critical tabular predictors (e.g., historical temperature, solar radiation), while Holt-Winters emphasizes temporal patterns such as daily seasonality and trend shifts, providing complementary insights for accurate weather forecasting.

2. Methodology

The methodology outlines the process for developing and evaluating the weather prediction models. Three models are employed: a Random Forest Regression (RFR) model for hourly predictions, a Holt-Winters Exponential Smoothing model for daily predictions, and a combined model to integrate the two predictions using shift and scale adjustments.

2.1. Data Collection and Preprocessing

Two distinct datasets were utilized in this study. The first dataset consists of historical daily temperature records (average, maximum, and minimum) from 1979 to 2018, used to train the Holt-Winters model for daily temperature forecasting. The second dataset comprises meteorological variables from May 2024, April 2025, and the first nine days of May 2025, including temperature, solar radiation, precipitation, and geographic coordinates. A feature matrix with 172,634 samples was constructed for training the Random Forest model, while 14,160 samples were reserved for prediction. All tabular data was standardized to ensure consistency.

2.2. Input Features

The Random Forest model utilizes the following input features, derived from the ERA5 dataset:

Table 1: Input Features for Random Forest Model

Feature	Description
latitude	Geographic latitude in degrees (e.g., 28.0000°N)
longitude	Geographic longitude in degrees (e.g., 120.7000°E)
d2m celsius	Dewpoint temperature at 2m in °C
ssr	Surface net solar radiation (W/m ²)
sshf	Surface sensible heat flux (W/m ²)
sp	Surface pressure (hPa)
tp	Total precipitation (mm)
hour sin	Sine of hour angle for diurnal periodicity
hour cos	Cosine of hour angle for diurnal periodicity
day of year sin	Sine of day-of-year angle for annual periodicity
day of year cos	Cosine of day-of-year angle for annual periodicity
t2m lag1	Lagged 2m temperature from previous hour (°C)
tp lag1	Lagged precipitation from previous hour (mm)

2.3. Mathematical Interpretation of the Models

2.3.1. Random Forest Regression for Hourly Prediction

The Random Forest Regression (RFR) model predicts hourly temperatures based on meteorological features, using an ensemble of decision trees to capture complex relationships in the data. The process is as follows:

- **Step 1: Bootstrap Sampling for Each Tree**

For each of the N trees, randomly sample a subset of the training data with replacement (bootstrap sampling). Specifically, select K samples and $\sqrt{\text{total features}}$ to introduce data-level randomness and ensure diversity among the trees.

- **Step 2: Build a Regression Tree on Each Subset**

Each tree is trained independently using the standard regression tree construction process:

1. Enumerate all features X_j and candidate split points s .
2. Split the data into left and right regions:

$$\mathbf{R}_1(j, s) = \{x \mid x_j \leq s\}, \quad \mathbf{R}_2(j, s) = \{x \mid x_j > s\}$$

3. Compute the total Mean Squared Error (MSE) for the split:

$$\text{MSE}(s) = \sum_{x_i \in \mathbf{R}_1} (y_i - \bar{y}_{R_1})^2 + \sum_{x_i \in \mathbf{R}_2} (y_i - \bar{y}_{R_2})^2$$

where $\bar{y}_{R_1}$ and $\bar{y}_{R_2}$ are the mean values of the target variable in the left and right regions, respectively.

4. Choose the split point (j, s) that minimizes the MSE, and recursively split until a stopping criterion is met.

- **Step 3: Construct N Trees**

Repeat the process N times to construct N regression trees:

$$T_1(x), T_2(x), \dots, T_N(x)$$

Each T_i is a regression tree trained on a different random subset of the data.

- **Step 4: Predict by Averaging**

For a new input x , each tree outputs a prediction $\hat{y}_i = T_i(x)$. The final predicted hourly temperature is the average of all tree outputs:

$$\hat{y}_{\text{hourly}} = \frac{1}{N} \sum_{i=1}^N T_i(x)$$

2.3.2. Seasonal Decomposition and Holt-Winters Exponential Smoothing for Daily Prediction

The Holt-Winters Exponential Smoothing model predicts daily temperatures by capturing the level, trend, and seasonality in the historical 5-month temperature data. The process involves seasonal decomposition followed by exponential smoothing, with adjustments for trend extrapolation and seasonality correction.

- **Step 1: Seasonal Decomposition**

The historical daily temperature data M_t (e.g., average temperature) is decomposed into trend, seasonal, and residual components using an additive model with a period of 31 days (the length of May):

$$M_t = T_t + S_t + R_t$$

where T_t is the trend component, S_t is the seasonal component with period $s = 31$, and R_t is the residual. The detrended data is computed by removing the trend:

$$y_t = M_t - T_t$$

The seasonal component for May 14 ($S_{\text{may}_{14}}$) is the average detrended value for all May 14 dates across the years:

$$S_{\text{may}_{14}} = S_{13} = \frac{1}{n_{\text{all May 14}}} \sum (M_t - T_t)$$

where the index 13 corresponds to May 14 (0-based indexing).

- **Step 2: Holt-Winters Exponential Smoothing**

The detrended data y_t is used to train an additive Holt-Winters model with a seasonal period $s = 31$. The model updates three components: level L_t , trend b_t , and seasonality S_t .

- **Initialization**

Initialize the components:

- * Level L_0 : Set as the average of the first cycle of y_t .
- * Trend b_0 : Set to 0 or the initial slope of y_t .
- * Seasonal indices $S_{-30}, \dots, S_{-1}$: Set to 0 or computed from the first cycle.

- **Update Equations**

For each time step t , update the components:

$$L_t = \alpha(y_t - S_{t-s}) + (1 - \alpha)(L_{t-1} + b_{t-1})$$

$$b_t = \beta(L_t - L_{t-1}) + (1 - \beta)b_{t-1}$$

$$S_t = \gamma(y_t - L_t) + (1 - \gamma)S_{t-s}$$

where α , β , and γ are smoothing parameters in $[0, 1]$, optimized by minimizing the mean squared error (MSE) during training.

– **Forecast**

To predict m days ahead (e.g., $m = 13$ for May 14 from May 1), the forecast is:

$$\hat{y}_{t+m} = L_t + mb_t + S_{t+m-s}$$

where S_{t+m-s} is the seasonal component from the previous cycle's corresponding day (e.g., for May 14, 2025, $S_{t+13-s} = S_{t-18}$, typically the last cycle's S_{13}).

• **Step 3: Trend Extrapolation**

Since $\hat{y}_{t+m}$ predicts the detrended temperature, the original trend T_t must be added back. The trend is extrapolated to 2025 using the last trend value T_{last} (from 2018-05-31) and the average trend difference:

$$\text{trend}_{\text{diff}} = \text{mean} \left(\frac{T_t - T_{t-1}}{\Delta t} \right) \times 0.5$$

$$\text{trend}_{\text{forecast}_i} = T_{\text{last}} + i \times \text{trend}_{\text{diff}}, \quad i = 0, 1, \dots, 30$$

For May 14, $i = 13$.

• **Step 4: Final Daily Prediction with Seasonal Adjustment**

The final daily prediction combines the Holt-Winters forecast, the extrapolated trend, and the historical seasonal component:

$$F_{\text{may}} = \hat{y}_{t+m} + \text{trend}_{\text{forecast}_i} + S_{\text{may}_{14}}$$

This ensures the prediction accounts for the long-term trend and the true seasonal pattern of May 14.

• **Step 5: Final Adjustment with Historical Mean and Fluctuation**

The prediction is adjusted using the historical mean of May temperatures:

$$\text{historical}_{\text{may mean}} = \frac{1}{N} \sum_t M_t$$

$$F_{\text{final}} = 0.9 \times \text{historical}_{\text{may mean}} + 0.1 \times F_{\text{may}}$$

Random fluctuation is added to simulate natural variability:

$$\text{may}_{\text{fluctuation}} \sim \mathcal{N}(0, 1.5)$$

$$\text{may}_{\text{forecast}} = F_{\text{final}} + \text{may}_{\text{fluctuation}}$$

2.3.3. Combined Model with Shift and Scale Adjustment

The hourly predictions from the Random Forest model and the daily predictions from the Holt-Winters model are combined to produce a final hourly forecast that aligns with the daily predictions. This is achieved through shift and scale adjustments.

• **Step 1: Shift Adjustment**

Compute the difference between the daily average (from Holt-Winters) and the hourly average (from Random Forest) for the same day:

$$\text{shift} = \text{daily}_{\text{avg}} - \text{hourly}_{\text{avg}}$$

Adjust the hourly predictions by adding the shift:

$$\text{adjusted}_{\text{temp}} = \text{pred}_{\text{temp}} + \text{shift}$$

This ensures the hourly predictions align with the daily average.

• **Step 2: Scale Adjustment**

Scale the adjusted hourly temperatures to match the daily range (maximum and minimum) predicted by Holt-Winters:

$$\text{current}_{\text{range}} = \text{adjusted}_{\text{temp.max}} - \text{adjusted}_{\text{temp.min}}$$

$$\text{target}_{\text{range}} = \text{daily}_{\text{max}} - \text{daily}_{\text{min}}$$

$$\text{scale} = \frac{\text{target}_{\text{range}}}{\text{current}_{\text{range}}}$$

$$\text{adjusted}_{\text{temp}} = \text{daily}_{\text{min}} + (\text{adjusted}_{\text{temp}} - \text{adjusted}_{\text{temp.min}}) \times \text{scale}$$

This ensures the hourly predictions have the same range as the daily predictions, preserving the daily variability.

2.3.4. Evaluation

The RFR model was evaluated on a validation set using:

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}, \quad (1)$$

$$\text{MAE} = \frac{1}{n} \sum |y_i - \hat{y}_i|, \quad (2)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum (y_i - \hat{y}_i)^2}, \quad (3)$$

where y_i , $\hat{y}_i$, and $\bar{y}$ are observed, predicted, and mean temperatures, respectively.

3. Results and Discussion

3.1. Random Forest Performance

Due to the large number of leaf nodes and the depth of the full decision tree, we only visualize the first three levels of its structure for clarity. Figure 1 illustrates the top layers of a single decision tree extracted from the trained Random Forest model. This partial view highlights the early-stage feature splits and their corresponding decision thresholds.

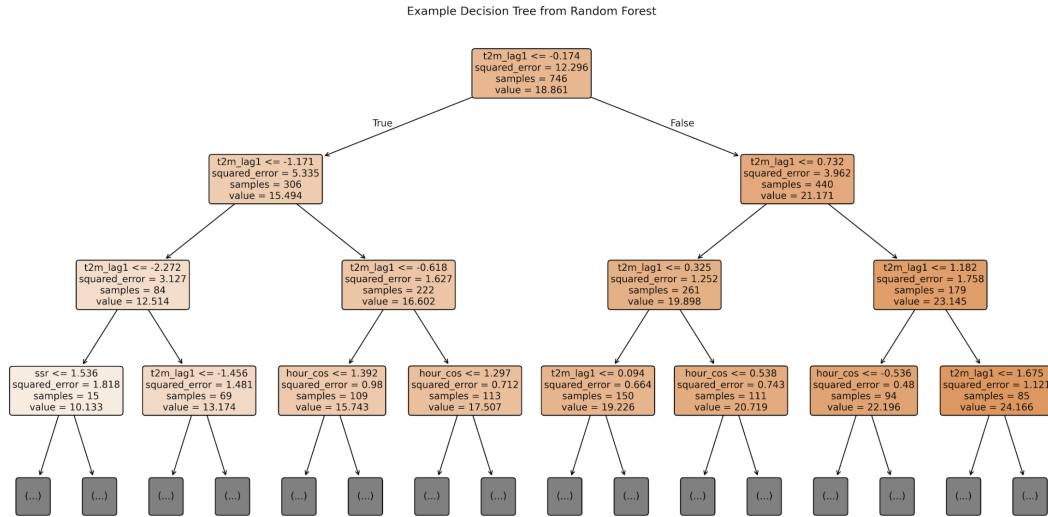


Fig. 1: Top three layers of a single decision tree from the trained Random Forest model.

- R^2 : 0.94, indicating a near-perfect fit to the training data, though suggesting potential overfitting.
- **Mean Absolute Error (MAE)**: 0.18, demonstrating high prediction accuracy for temperature.
- **Root Mean Square Error (RMSE)**: 0.26, reflecting low error variance.

3.2. Seasonal Decomposition and Holt-Winters Exponential Smoothing for Optimization

To evaluate the effectiveness of Holt-Winters optimization, we compare the original Random Forest hourly predictions with the adjusted results after applying seasonal smoothing. Figure 2 below displays the predicted temperature curves before and after optimization.

For external validation, Figure 3 shows actual hourly temperature data on May 14, 2025, retrieved from AccuWeather. This comparison helps assess how well the optimized forecast aligns with real-world observations.

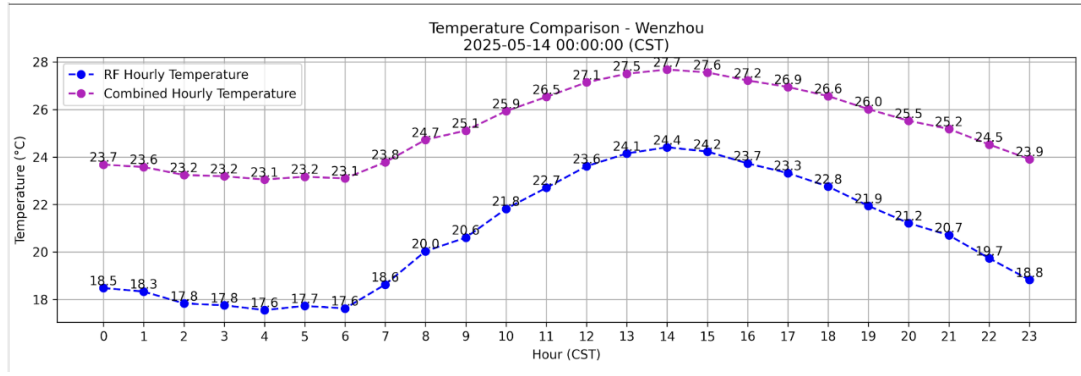


Fig. 2: Hourly forecast before and after Holt-Winters optimization.

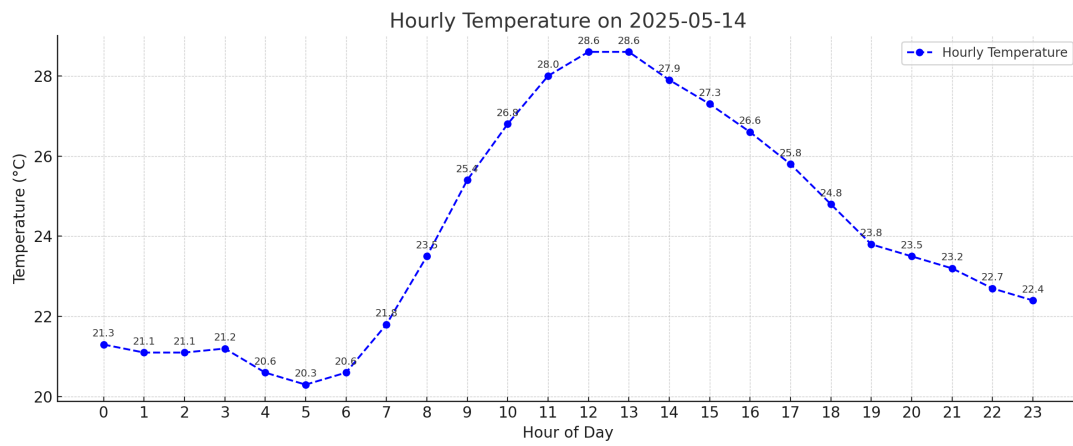


Fig. 3: Actual hourly temperature from AccuWeather (May 14, 2025).

4. Limitations of Weather Prediction Models

- **Lack of temporal dependence in RFR:** The Random Forest model predicts each hour independently without capturing sequential dependencies, which may lead to less smooth temperature curves.
- **Post-hoc nature of Holt-Winters:** The Holt-Winters smoothing is applied after the Random Forest predictions, meaning it cannot influence the learning process or adjust the model structure directly.
- **Dependence on historical patterns:** The optimization process relies heavily on past seasonal averages (e.g., historical May trends), which may suppress legitimate anomalies in future forecasts.

References

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